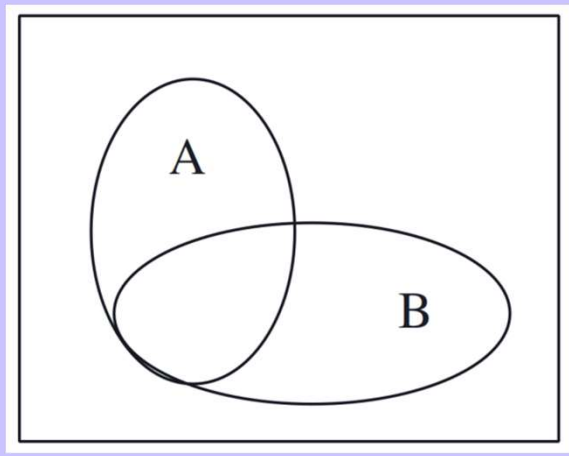
USES OF CONDITIONAL PROBABILITY

The Product Rule, Bayes' Rule, and Extended Independence

Today's Agenda

- Recall: Conditional Probability
- Example of Multiplication Rule
- Total Probability Theorem
  - Intuition
  - Definition
  - Examples
- Bayes' Rule

Conditional Probability: Definition



- $\mathbf{P}(A \mid B)$ = probability of A, given that B occurred
  - B is our new universe
- **Definition:** Assuming $\mathbf{P}(B) \neq 0$,

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$\mathbf{P}(A \mid B)$ is undefined if $\mathbf{P}(B) = 0$

The Product Rule

- Notation: In advanced probability courses it is common to denote the intersection of events by concatenating them rather than writing an intersection symbol between them. That is, if A and B are events we write AB instead of $A \cap B$. Our book does not use this notation, but I will use it in this lecture to simplify what I have to type (i.e., writing the letters adjacent does not make me use the equation editor!).

The Product Rule

- Example: Suppose a bag contains four green beads and seven red ones. If I pull two out (without replacement), what is the probability that both are red?
  - Intuitively my probability of getting red on the first draw is 7/11, my probability of *then* getting another red on the second draw is 6/10=3/5, and my total probability of getting both red is (7/11)*(3/5)=21/55, which is about 0.38.
  - Formally we can solve the problem by counting. There are C(11,2)=55 2-subsets of the 11 beads, and C(7,2)=21 of them contain 2 red beads. Assuming all outcomes are equally likely, the probability of 2 reds is 21/55. We see that our intuitive procedure yields the correct answer, but can we justify it formally?

The Product Rule

- Theorem (The Product Rule for Probabilities): Suppose A and B are events in a sample space S with probability measure P. We already know that P(B|A)=P(AB)/P(A). Clearing denominators we see P(AB)=P(A)P(B|A). In other words, the probability that A and B both happen equals the probability that A happens times the probability that B happens if A does.

Multiplication Rule

Assuming that all of the conditioning events have positive probability, we have

$$\mathbf{P}\left(\cap_{i=1}^{n} A_i\right) = \mathbf{P}(A_1)\mathbf{P}(A_2 \mid A_1)\mathbf{P}(A_3 \mid A_1 \cap A_2)\cdots\mathbf{P}\left(A_n \mid \cap_{i=1}^{n-1} A_i\right).$$

The Product Rule

- Example Revisited: Again suppose we have four green and seven red beads in a bag and we choose two of them without replacement. Let F be the event that the first bead chosen is red and S be the event that the second bead chosen is red. Then the even that both beads are red is FS. We can find its probability from the product rule as follows: $P(FS)=P(F)P(S|F)=(7/11)*(6/10)=21/55$. Our intuition is now justified.

The Product Rule

- The product rule generalizes to more events. For instance, suppose A, B, C, and D are events in a sample space S with probability measure P. Then P(ABCD)=P(A)P(B|A)P(C|AB)P(D|ABC). This same pattern works with any number of events.
- Example Re-revisited: Now suppose we have four green and seven red beads and we want the probability that four beads chosen (without replacement) from the bag are all red. Let A, B, C, and D, be the events that the first, second, third, and fourth beads are red, respectively. Then the probability that all four beads are red is P(ABCD)=(7/11)*(6/10)*(5/9)*(4/8) =(7/11)*(3/5)*(5/9)*(1/2)=7/66, which is about 0.11. Note that, for instance, the fourth factor 4/8=1/2 is the probability that the fourth bead is red after we have already removed three red beads.

Independence of three events

- If A, B, and C are events in some sample space S with a probability measure P, then we say the events are independent if each pair is independent (i.e., $P(AB)=P(A)P(B)$, $P(AC)=P(A)P(C)$, and $P(BC)=P(B)P(C)$) and in addition $P(ABC)=P(A)P(B)P(C)$.
- Intuitively this is what is needed to guarantee that all the events and their complements are proportionally represented in each other in all relevant ways. It is possible to satisfy some of these equations while failing to satisfy others

Independence of three events

- Contrary Example: Roll a red die and a clear die. Let A be the event "the red die is 1," B be the event "the clear die is 3," and C be the event "both dice have the same number." The P(A)=P(B)=P(C)=6/36=1/6. Clearly P(AB)=1/36=P(A)P(B), P(AC)=1/36=P(A)P(C), and P(BC)=1/36=P(B)P(C). The event ABC, however, is empty (since you cannot have the red die 1, the clear die 3, and both dice the same), so P(ABC)=0. But P(A)P(B)P(C)=(1/6)(1/6)(1/6)=1/216. Thus A, B, and C are not independent.

Independence of three events

- Example: Roll red, green, and clear dice. Let A be the event “the red dice is 1,” B be the events “the green die is 1,” and C be the event “the clear die is 1.” It is easy to test that events A, B, and C are independent under the uniform model. In particular $P(ABC)=1/216=(1/6)(1/6)(1/6)=P(A)P(B)P(C)$.

Independence of more events.

- A collection of events is independent if the probability of every subset of them equals the product of the probabilities of the events in the subsets. For example, events A, B, C, and D are independent if every pair is independent, every triple is independent, and P(ABCD)=P(A)P(B)P(C)P(D). (Consider rolling four dice and getting all 1's, flipping four coins and getting HTTH).

Bayes' Rule

- Bayes' Rule is a simple formula relating the values of P(A|B) and P(B|A). It has several forms and interesting consequences.

Bayes' Rule

Let $A_1, A_2, \ldots, A_n$ be disjoint events that form a partition of the sample space, and assume that $\mathbf{P}(A_i) > 0$, for all i. Then, for any event B such that $\mathbf{P}(B) > 0$, we have

$$\mathbf{P}(A_i \mid B) = \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\mathbf{P}(B)}$$

$$= \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \cdots + \mathbf{P}(A_n)\mathbf{P}(B \mid A_n)}.$$

Bayes' Rule

- Theorem 2.16 (Bayes' Rule)
  - Given events H and E in a sample space S with probability measure P it holds that $P(H|E)=P(H)P(E|H)/P(E)$
  - Proof: By definition of conditional probability, $P(H|E)=P(HE)/P(E)$. By the product rule $P(HE)=P(H)P(E|H)$. Therefore $P(H|E)=P(H)P(E|H)/P(E)$.
  - Here we use H and E to stand for Hypothesis and Evidence. We sometimes conceive of Bayes' Rule as telling us how to revise the probability of a hypothesis based on the observation of some particular piece of evidence.

Bayes' Rule

- Example: You are living in a dorm. One night the fire alarm goes off. How likely is it that there is a fire? Here H is the event "there is a fire" and E is the event "the fire alarm goes off." You want to know P(H|E). You estimate that all things being equal a fire is unlikely on a given night, setting P(H)=0.001 (roughly one fire in three years). You know that in a typical semester of about 100 days there are about 3 fire alarms (typically false alarms), so you estimate P(E)=0.03. Finally you guess that it is nearly certain someone would set off the alarm if there really were a fire, so you estimate P(E|H)=0.98. By Bayes' Rule, P(H|E)=P(H)P(E|H)/P(E)=(0.001)(0.98)/(0.03)=0.033.

Bayes' Rule

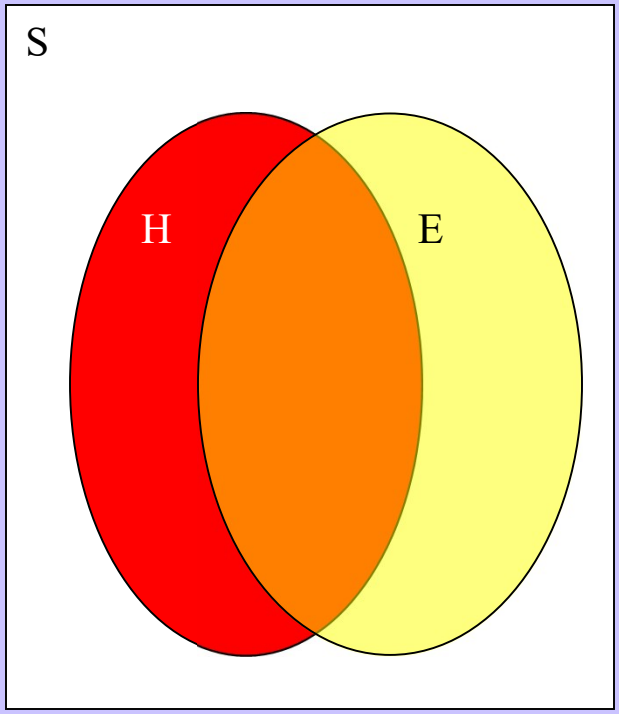
- Notes to the example: From one point of view the alarm is almost meaningless. There is only 3.3% chance of a fire. Why is it so low? Your probabilities say that in 100 days you should expect 30 alarms but only one fire. Thus your chance of having fire with the alarm is 1/30. From another point of view the alarm carries a lot of weight: The alarm raises the likelihood of a fire thirtyfold, from 0.1% to 3.3% (that is 1/1000 to 1/30). This is how the evidence (alarm) causes you to revise your estimate of the hypothesis (fire). In any case the difference between P(H|E) and P(E|H) is large: 0.033 to 0.98, a clear example of how these quantities need not be equal.

Bayes' Rule

- Theorem 2.17 (Bayes' Rule, extended form)
  - Under the same circumstances as before .

$$P(H \mid E) = \frac{P(H)P(E \mid H)}{P(H)P(E \mid H) + P(\overline{H})P(E \mid \overline{H})}$$

Bayes' Rule



- Proof: This is the same equation as in the simpler statement except that in that case the denominator was simply P(E). It is easy to see that the sets EH and $E\overline{H}$ partition E. The Venn Diagram makes it clear: The event E is partitioned into the yellow section EH and the orange section $E\overline{H}$. Since the sets are disjoint and have union E, we have $P(E) = P(EH) + P(E\overline{H})$ By the product rule the righthand side becomes $P(H)P(E|H) + P(\overline{H})P(E|\overline{H})$ So we have the same equation as before, but with a fancy expansion of P(E) in the denominator.

Bayes' Rule

- Example (medical testing)
  - A drug company has designed a test for a disease. Through extensive testing, the company reports that the test produces only 1% false positive results (i.e., a healthy person tests positive) and only 2% false negative results (i.e., a person with the disease tests negative). Let P be the event "someone tests positive," N be the event "someone tests negative," H be the event "someone is healthy," and D be the event "someone has the disease." Then the company is reporting P(P|H)=0.01 (or equivalently P(N|H)=0.99) and P(N|D)=0.02 (or equivalently P(P|D)=0.98).

Bayes' Rule

- Example (medical testing)
  - Suppose you test positive for the disease. How likely is it that you in fact have the disease? It is tempting but incorrect to say 98% since P(P|D)=0.98. But you want to know P(D|P), which may be quite different. It turns out you do not have enough information yet. Oddly enough you must also know P(D), the prevalence of the disease in your population. Why? As in the case of the fires and fire alarms, if the disease is rare, then false positives will dominate true ones. If the disease is common, true positives will dominate false ones.

Bayes' Rule

- Example (medical testing)
  - Suppose the disease is rare, occurring in only 0.05% of the population. Then applying the second form of Bayes' we get

$$P(D \mid P) = \frac{P(D)P(P \mid D)}{P(D)P(P \mid D) + P(H)P(P \mid H)}$$

$$= \frac{0.0005 * 0.98}{0.0005 * 0.98 + 0.9995 * 0.01} \approx 0.047 = 4.7\%$$

Bayes' Rule

- Thus with a positive test your chance of having the disease is still just below 5%. Why? Roughly speaking among 2000 randomly chosen people you expect to have 20 positive tests but only 1 person with the disease. Thus about 95% of your positives are false. Still this is a dramatic increase in the probability of having the disease, from 0.05% to 4.7%, almost a hundredfold increase.

Bayes' Rule

- On the other hand, suppose the disease is common. Suppose 10% of people in your "population" have the disease. Then

$$P(D \mid P) = \frac{P(D)P(P \mid D)}{P(D)P(P \mid D) + P(H)P(P \mid H)}$$

$$= \frac{0.1 * 0.98}{0.1 * 0.98 + 0.9 * 0.01} \approx 0.916 = 91.6\%$$